

Hierarchy of Generalized Contractions

Banach

$$d(Tx, Ty) \leq k \cdot d(x, y)$$

Kannan

$$d(Tx, Ty) \leq r[d(x, Tx) + d(y, Ty)]$$

Chatterjea

$$d(Tx, Ty) \leq r[d(x, Ty) + d(y, Tx)]$$

Zamfirescu

$$d(Tx, Ty) \leq \max \{d(x, y), \frac{1}{2}[d(x, Tx) + d(y, Ty)], \frac{1}{2}[d(x, Ty) + d(y, Tx)]\}$$

Ćirić (HR)

$$d(Tx, Ty) \leq a_1 d(x, Tx) + a_2 d(y, Ty) + a_3 d(x, Ty) + a_4 d(y, Tx) + a_5 d(x, y)$$

H-S General

$$d(Tx, Ty) \leq \phi(d(x, Tx), d(y, Ty), d(x, Ty), d(y, Tx), d(x, y))$$